

An Assessment of GPT-Prompting for Labeling Text in the Context of Virtue Ethics



Evelin Buestan, '26, Lucy Engels, '27, Artemist Pierre '27, Yuhe Xue '27
Advised by Prof. Eni Mustafaraj (Computer Science)

Background

- This poster explores using GPT prompting to label texts of three contextual datasets that are analyzed in the companion poster, *A Computational Examination of Virtue Ethics Presence in Three Textual Datasets*.
- Due to the large number of textual units in each dataset, **manual labeling would have been costly and time consuming**. Applying GPT made what would have taken days of work of manual labeling be completed in a much more efficient way, within hours. However, finding the right prompt is a laborious process.
- GPT Prompting** refers to interacting with a Generative Pre-trained Transformer (GPT) by **providing it with specified instructions** to perform certain tasks and return desired responses.
- Model GPT - 4o** was used to label text in this study.

How to Measure Text

To compare text units, we use GPT to assign ordinal labels (scores) according to various scales, as described below.

Wellesley Course Catalog: GPT is prompted to give a score on a 5-point scale evaluating the extent to which each of Shannon Vallor’s techno moral virtues is present in the text. A score of **1** indicates that the **virtue is minimally present** in the course and **5** indicates that it is **very present**.

Wellesley News: A 5-point scale was used to measure the relevance of each article to two dimensions, “technology and its use” and “ethics and societal impact”. A score of **1** means “**minimally mentions**”, and a score of **5** means “**heavily mentions**”. A score of 4 or 5 was then defined as an indication of the high relevance of the article to the dimension the score measures

Sidechat: A 3-point scale was used to measure two dimensions, **Tone** and **Content**. The Tone is scored from **-1 to 1**, where -1 means being **rude** and 1 means being **polite**. The Content also use the same scale, where -1 means the **opposite of Shannon Vallor’s technomoral virtues**, and 1 means **displaying the virtue fully**.

Evaluating GPT Results

Validation Test

The Cohen’s Kappa is used to measure the agreement between the results of GPT labeling and Human labeling.

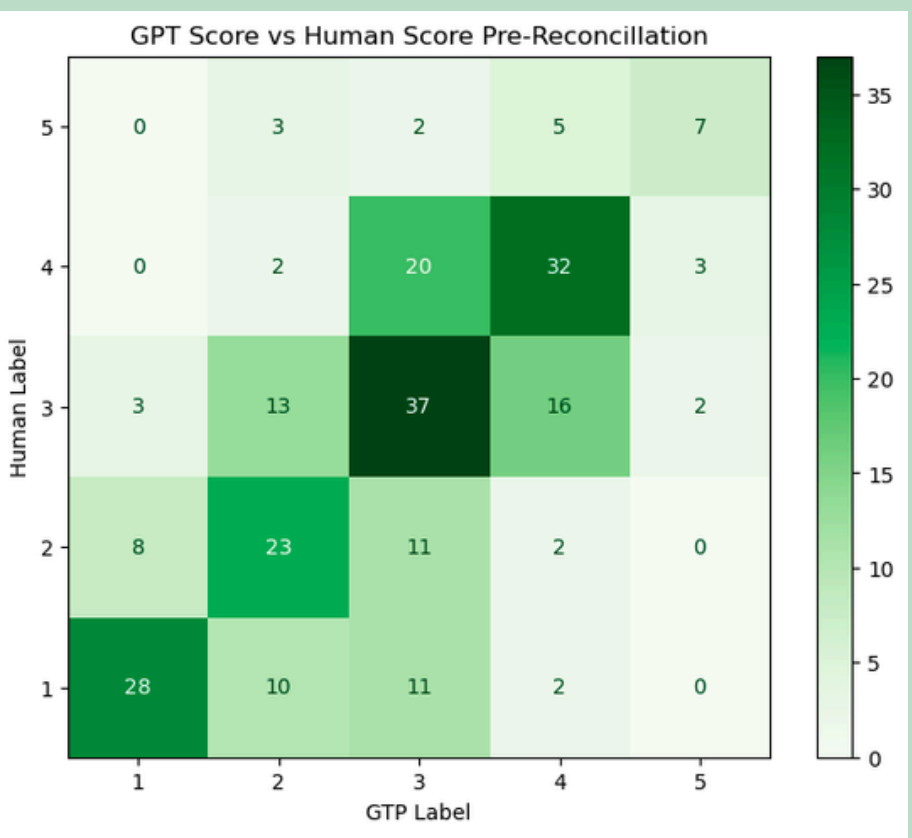
Cohen’s Kappa (K)

$$K = \frac{(\text{observed agreement} - \text{expected agreement})}{(1 - \text{expected agreement})}$$

- $\kappa < 0$: Less than chance agreement
- $\kappa = 0 - 0.20$: Slight agreement
- $\kappa = 0.21 - 0.40$: Fair agreement
- $\kappa = 0.41 - 0.60$: Moderate agreement
- $\kappa = 0.61 - 0.80$: Substantial agreement
- $\kappa = 0.81 - 1.00$: Almost perfect agreement

Course Catalog

- Used GPT to examine how much a course description focused on each of Shannon Vallor’s technomoral virtues
- It takes **2-3 minutes for a human** to evaluate a single course without a justification, and **3-5 minutes** with a justification
- Takes **15-20 seconds for AI** to label a course with justifications
- Prompt provided a rubric to use when scoring and two examples, using the “few shot” technique of prompting
- Used a **sample of 20 courses** to evaluate agreement

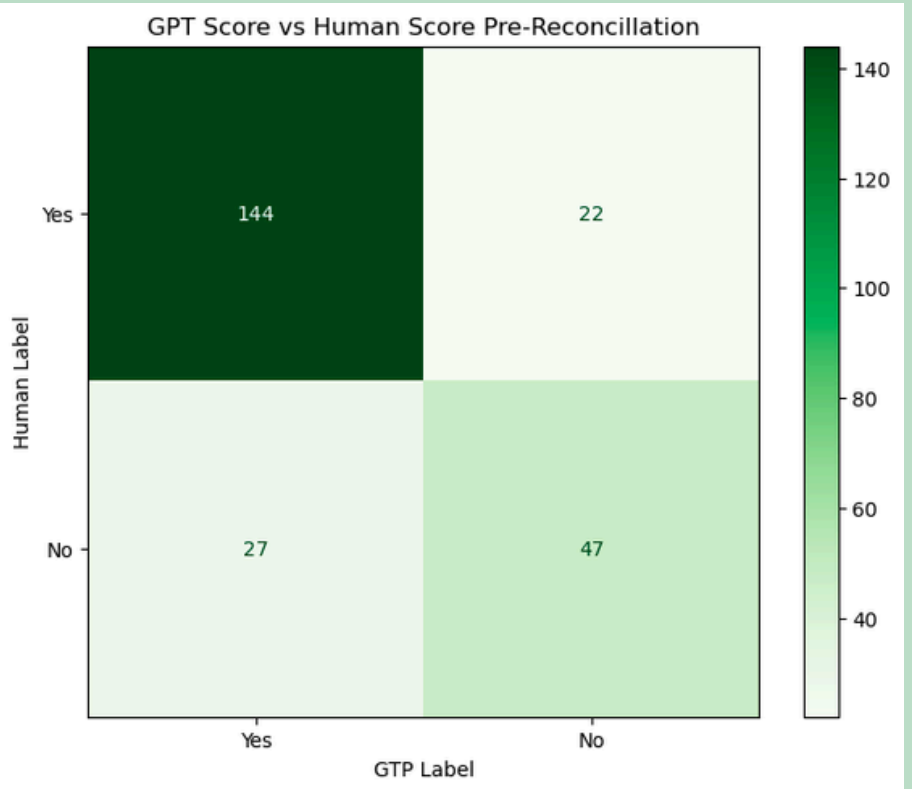


0.38

5-point agreement
52% (127 / 240)
exact agreement
89% (213/240)
within 1 point

0.51

80% (191 / 240)
Binary agreement



GPT Example: Disagree

Course: CPLT 294: Utopia and Dystopia in Literature

Virtue: Civility **GPT Ranking:** 3 **Human Ranking:** 5

GPT Justification:

Discussions around utopian and dystopian societies often involve collective deliberation about societal norms and responsibilities, promoting civility and respect.

GPT Example: Agree

Course: ARTS 321/ CAMS 321 - Advanced New Media

Virtue: Empathy **GPT Ranking:** 4 **Human Ranking:** 4

GPT Justification:

Collaborative projects and peer critiques in the course promote empathy, as understanding others' perspectives is necessary for effective collaboration and constructive feedback.

Discussion

Course Catalog:

- While most GPT justifications lined up with a human’s understanding, there were some places where a human’s interpretation significantly differed from GPT’s understanding
- Creating a successful prompt required many iterations. Reaching a Cohen’s Kappa score of 0.38 required 14 attempts of iterating across the prompt and the rubric
- There is a level of subjectivity, as illustrated in the example where GPT and a human rater disagreed.

Wellesley News:

- GPT tends to generate more accurate scores when the prompt asks it to include justifications for its scores.
- GPT also generates responses with lower variability when the prompt specifies a narrow range of options for the desired response. For example, question “answer whether the article is subjective or objective” gives either “subjective” or “objective” as the answer, while question “answer whether the article is subjective or objective” may lead to various unexpected answers such as “NA” or “objective and neutral”.
- Clear and concise prompts make GPT generate best responses, yet what is clear and concise enough to GPT could only be realized through trial and error.

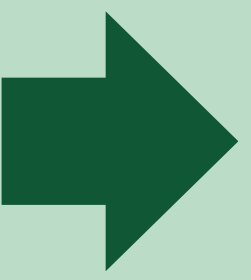
Sidechat:

- GPT scored the **contents** of **34** posts as unvirtuous while the human labeler scored **13** as such.
- GPT is more likely than the individual to score a post as unvirtuous.
- This is partially due to GPT’s **misinterpretations of emojis and slang**.
- The Cohen’s Kappas however show substantial agreement so using **GPT is a dependable choice to label Sidechat posts** and will be improved on.

Wellesley News

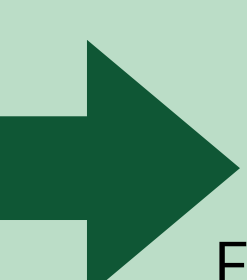
Prompt for GPT

```
message:
  • [{"role":"system","content":"You are a helpful assistant trained to give an answer to user's question. You will format the answer as a JSON object with keys Technology and Its Use, Ethics and Societal Impact, Subjective or Objective, Stance, Summary, and Reason for the Scores. Each JSON object should not be in a docstring"},
  • {"role":"user","content":"I need your help with summarizing a news article with respect to a few dimensions:
    ◦ to what extent it is about technology and its use (respond with a score from a 5-point scale from 1, representing not at all, to 5, representing entirely)
    ◦ to what extent it is about ethics or societal impact (respond with a score from a 5-point scale from 1, representing not at all, to 5, representing entirely)
    ◦ whether it's subjective or objective (respond one word out of subjective or objective)
    ◦ what is the stance of the author (if the article is subjective, respond one word out of approving, disapproving, or neutral; if the article is objective, respond NA)
    ◦ then provide a very short summary of the article (Respond to this in one string)
    ◦ then provide you reasoning for the scores you gave (Respond to this in one string). End of instructions. What follows is the text of the article:(text_to_analyze)"}]]
```



An example of an output from GPT that received 5 for both dimensions

original_articles	response
<pre>{ • "title": "Scientists Construct an Embryo Derived from Three Parents", • "authors": xxx • "date": xxx • "category": xxx • "url": xxx • "body": xxx }</pre>	<pre>{ • "Technology and Its Use": 5, • "Ethics and Societal Impact": 5, • "Subjective or Objective": "objective", • "Stance": "NA", • "Summary": xxx • "Reason for the Scores": xxx }</pre>



Validation test

- Human readers manually labeled a sample of 30 articles
- Cohen’s Kappa (k)

For “Technology and Its Use”:

0.491

For “Ethics and Societal Impact”:

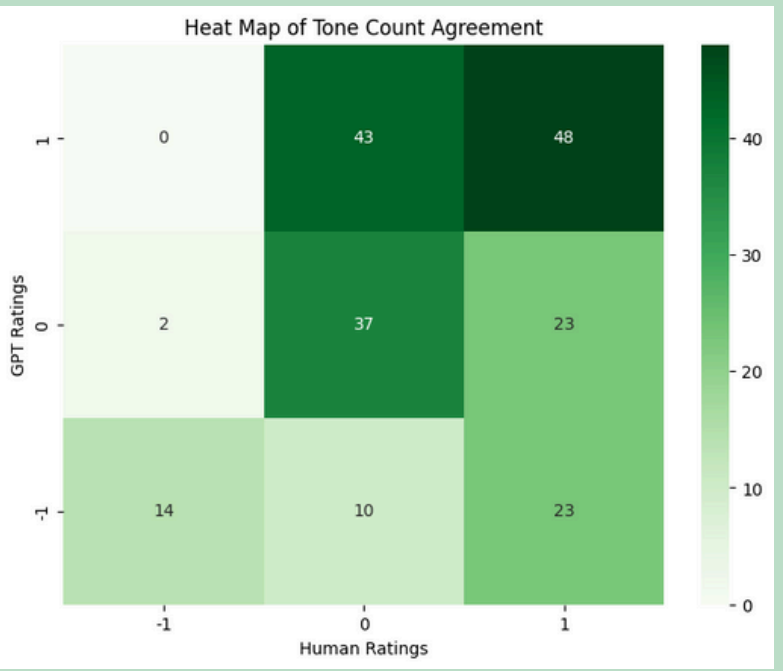
0.747



Conclusion: While GPT can be trusted to perform text labeling on Wellesley News, prompt could be further modified to enhance GPT’s agreements with human readers on scoring for “Technology and Its Use”

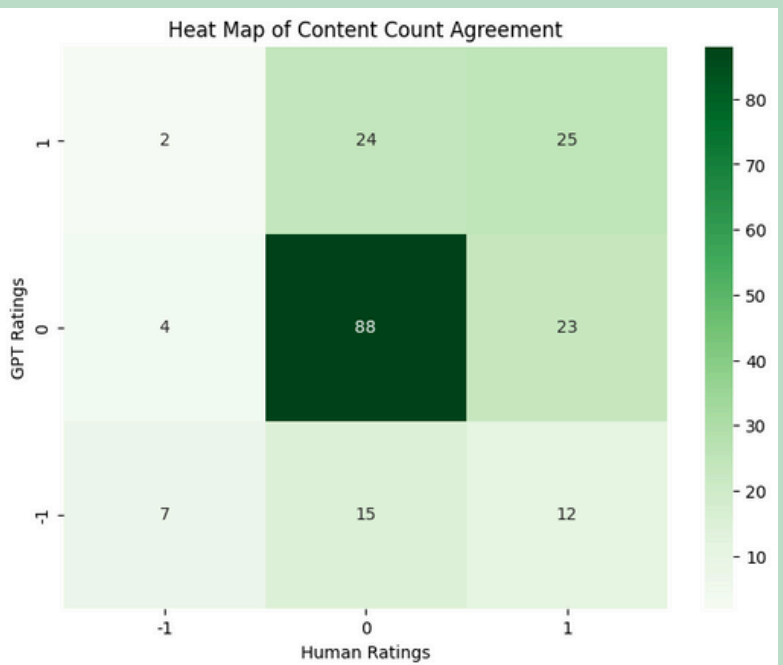
Sidechat

- Labeling 200 posts took a human **5 hours** while GPT took **2 minutes**.
- Kappa for **Tone** and **Content** show **significant agreement**
- Agreement for **unvirtuous** tone and content is **comparatively lower** than virtuous and neutral agreement
- GPT can **misinterpret** emojis and slang



0.76

Tone Agreement

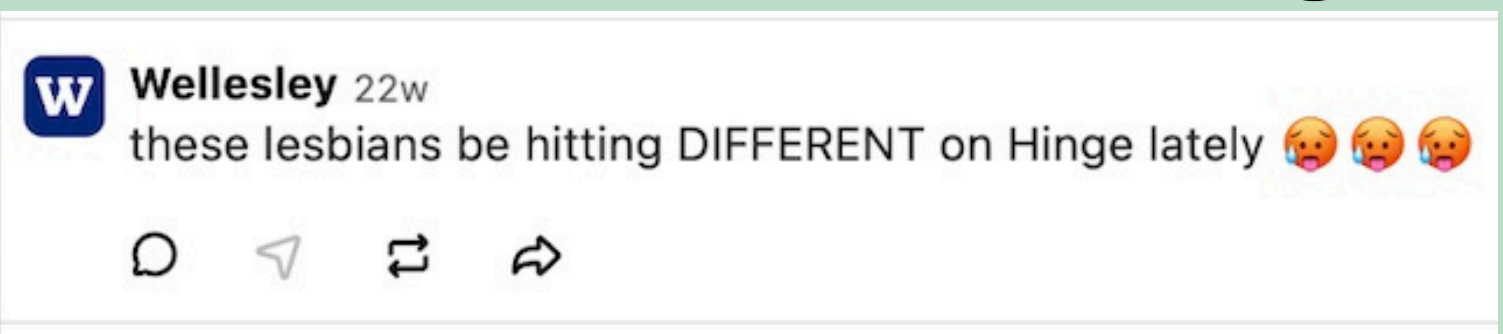


0.73

Content Agreement

Figure 2: Heat Map of the Tone and Content Agreement for 200 messages where -1 represents vices, 0 no virtue or vice present, and 1 represents virtues.

GPT’s Inaccurate Scoring



Tone: -1
Content: -1

Tone Score Reasoning: The tone is rude and dismissive, indicated by the use of negative emojis and an overall disrespectful attitude.

Content Score Reasoning: The message does not address any virtues and contains a general sense of negativity and perhaps implicit bias or judgment.

How would you score this?

Future Work

For all three datasets, we aim to:

- Improve GPT prompting by making the prompts more clear and concise in order for the responses to reach better agreement with human readers
- Take a step forward to explore GPT few-shot learning (providing a few labeled examples) or fine tuning (training with many labeled examples)
- Validate results with more than one human reader to see the inherent level of subjectivity of the tasks

Acknowledgement

Thank you to Prof. Eni Mustafaraj and team leader Johanna Lee for their care, support, and most of all patience. Thank you National Science Foundation and T.T. and W.F. Chao Summer Scholars Program in Natural Science for funding our research.